Skewed Datasets (Optional)

A **skewed dataset** is a classification problem where the ratio of positive (1) to negative (0) examples is very far from 50/50.

- **Example:** Rare disease detection, where only 0.5% of patients have the disease ($y = 1$) and 99.5% do not ($y = 0$).

The Problem with Classification Error

Using **classification error** (or **accuracy**) as your only metric is highly misleading for skewed datasets.

- **Example:** If the disease is present in only 0.5% of the population, a "dumb" (and useless) algorithm that *always* predicts $y = 0$ will have:
  - **0.5% error**
  - **99.5% accuracy**
- If your complex machine learning model achieves a 1% error (99% accuracy), it *looks* good, but it's performing *worse* than the useless algorithm that just guesses "no" every time.
- This shows that accuracy is not a good metric for telling if your model is actually useful.

Better Metrics: Precision and Recall

For skewed datasets, we use two different metrics: **Precision** and **Recall**. To calculate them, we first build a **Confusion Matrix**.

The Confusion Matrix

A confusion matrix is a 2x2 table that breaks down your model's predictions on a test set (or CV set) into four categories:

	Actual Class: 1 (Has Disease)	**Actual Class: 0 (No Disease)**
Predicted Class: 1	**True Positive (TP)** *Correctly predicted "yes"*	**False Positive (FP)** *Wrongly predicted "yes"*
Predicted Class: 0	**False Negative (FN)** *Wrongly predicted "no"*	**True Negative (TN)** *Correctly predicted "no"*

Precision

- **Question:** Of all the times the model predicted a patient **has the disease** ($y = 1$), what fraction of those patients **actually** had the disease?
- **Intuition:** "When my model says 'yes', how often is it right?" (This is a measure of *trustworthiness*).
- **Formula:**
$Precision = \frac{\text{True Positives}}{\text{Total Predicted Positives}} = \frac{TP}{TP+FP}$

Recall

- **Question:** Of all the patients who **actually had the disease**, what fraction did the model **correctly detect**?
- **Intuition:** "Of all the 'yes' cases that actually exist, how many did my model find?" (This is a measure of *completeness*).
- **Formula:**
$Recall = \frac{\text{True Positives}}{\text{Total Actual Positives}} = \frac{TP}{TP+FN}$

Using Precision and Recall

- The "dumb" algorithm that *always* predicts $y = 0$ would have 0 True Positives (TP).
- Its **Recall would be 0%** ($\frac{0}{0+FN}$).
- This metric immediately and correctly tells you the algorithm is useless.
- A good algorithm must achieve **both** high precision and high recall.